EKF for Battery System

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Abstract

This article will explain what and how the EKF is used and deployed to the actual drone with Ardupilot platform.

1 Battery Model

1.1 State

$$SOC(k+1) = SOC(k) + \frac{dt}{Q_{nom}}(I(k) + w(k)), \tag{1}$$

$$i_{bv}(k+1) = \frac{\tau}{\tau + \Delta t} i_{bv}(k) + \frac{\Delta t}{\tau + \Delta t}(I(k) + w(k)) \tag{2}$$

1.2 Output

$$v_{cell}(k) = OCV(SOC(k)) + R_{ac}I(k) + R_{bv}i_{bv}(k) + v(k) \tag{3}$$

2 EKF

2.1 Linearization

$$\begin{bmatrix} SOC(k+1) \\ i_{bw}(k+1) \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & \frac{\tau}{\tau+\Delta t} \end{bmatrix} \begin{bmatrix} SOC(k) \\ i_{bw}(k) \end{bmatrix} + \begin{bmatrix} \frac{dt}{Q_{nom}} \\ \frac{\Delta t}{\tau+\Delta t} \end{bmatrix} (I(k) + w(k)), \tag{4}$$

$$\begin{bmatrix} v_{cell}(k) \end{bmatrix} = \begin{bmatrix} \frac{dOCV}{dSOC}(k) & R_{bv} \end{bmatrix} \begin{bmatrix} SOC(k) \\ i_{bw}(k) \end{bmatrix} + \begin{bmatrix} R_{ac} \end{bmatrix} I(k) + v(k) \tag{5}$$

2.2 Error System

According to (4), and (5), let

$$x = \begin{bmatrix} SOC & i_{bv} \end{bmatrix}^T, \quad u = [I]^T, \quad y = [v_{cell}]^T \tag{6}$$

$$A = \begin{bmatrix} 1 & 0 \\ 0 & \frac{\tau}{\tau+\Delta t} \end{bmatrix} \tag{7}$$

$$B_u = B_w = \begin{bmatrix} \frac{dt}{Q_{nom}} \\ \frac{\Delta t}{\tau+\Delta t} \end{bmatrix} \tag{8}$$

$$C = \begin{bmatrix} \frac{dOCV}{dSOC} & R_{bv} \end{bmatrix} \tag{9}$$

$$D_u = \begin{bmatrix} R_{ac} \end{bmatrix}, \quad D_v = \begin{bmatrix} 1 \end{bmatrix} \tag{10}$$

System can be represent as this,

$$x(k+1) = Ax(k) + B_u u(k) + B_w w(k), \tag{11}$$
$$y(k) = Cx(k) + D_u u(k) + D_v v(k) \tag{12}$$

The estimated model is

$$\hat{x}(k+1) = A\hat{x}(k) + B_u u(k), \tag{13}$$
$$\hat{y}(k) = C\hat{x}(k) + D_u u(k) \tag{14}$$

Then the error system should be,

$$\tilde{x}(k+1) = A\tilde{x}(k) + B_w w(k), \tag{15}$$
$$\tilde{y}(k) = C\tilde{x}(k) + D_v v(k) \tag{16}$$

This should be the system for EKF.

3 Result

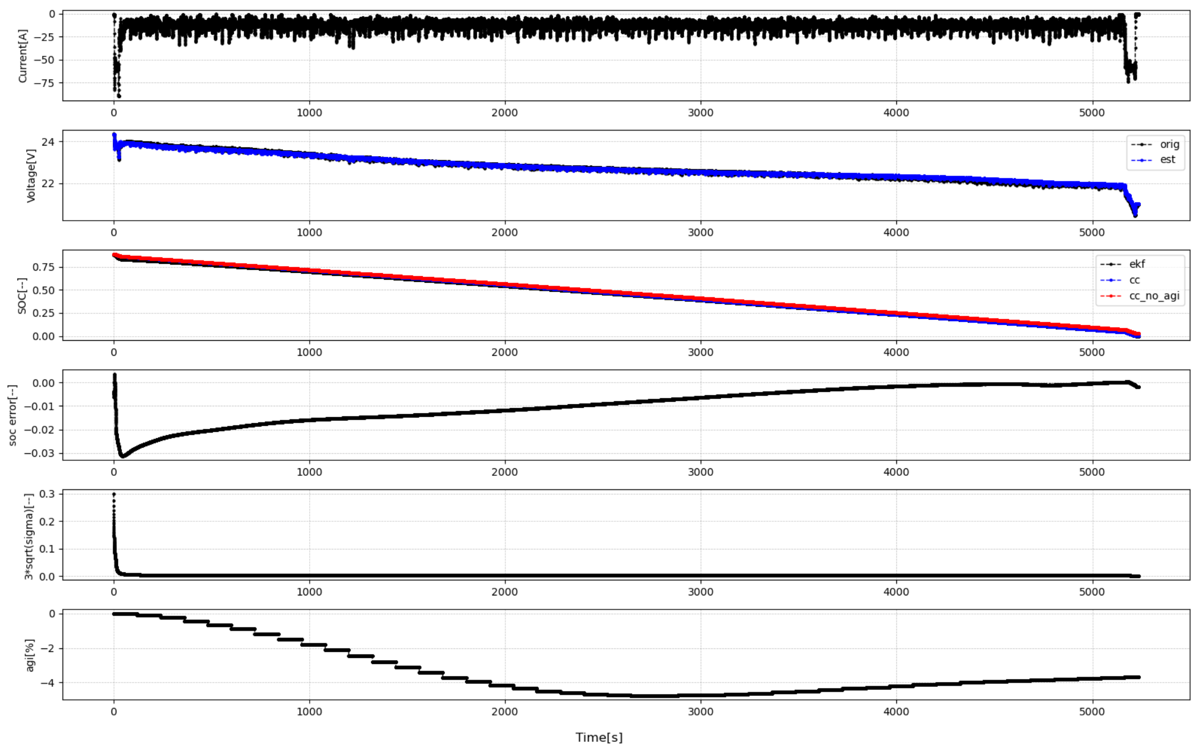

Figure 1: Result of flight test.

Figure 1 is the full trajectory of current, voltage, SOC, and SOH. The battery is around 3.2% less than the total capacity tested (21.671 Ah). The EKF estimate SOC with the 21.671 Ah and manage to track the actual SOC shown in blue.

Figure 2 shows the convergence of the error $\tilde{x} = x - \hat{x}$. The $3\sqrt{\sigma(\tilde{x})} \leq 0.02$ after 20 seconds, which shows the confidence of the estimation.

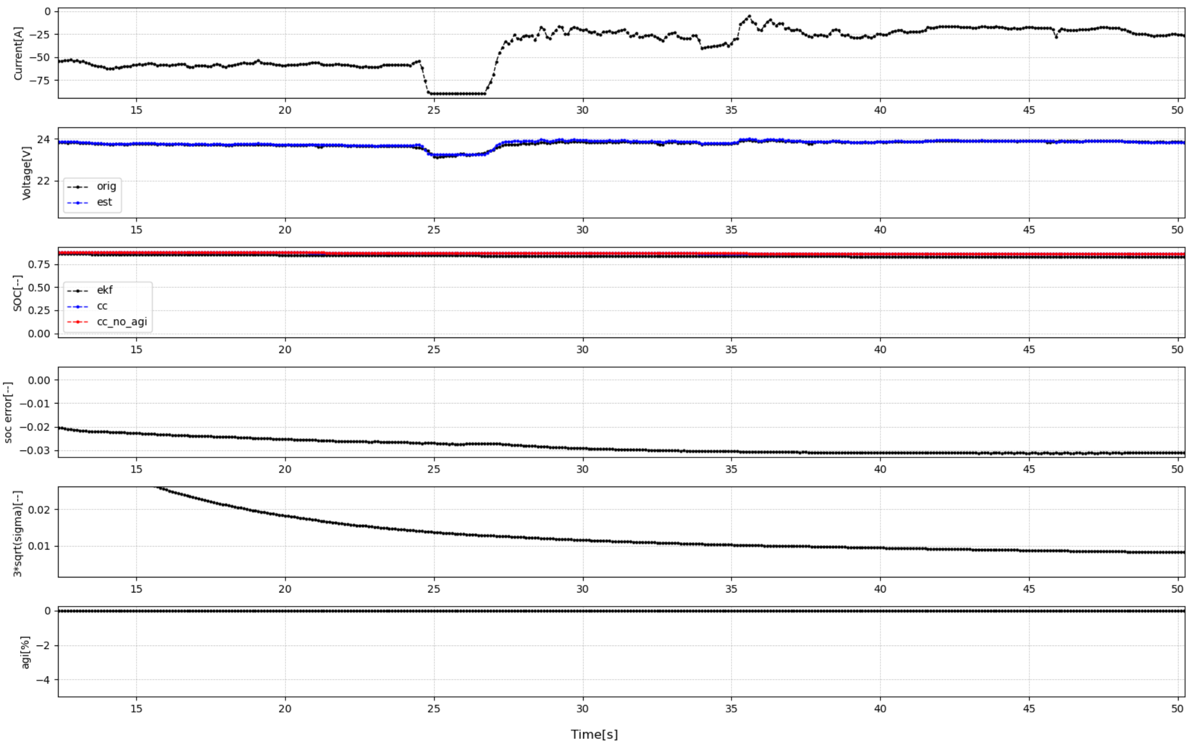

Figure 2: Convergence of sigma value for SOC estimation error.